Supplementary material for "An Infrastructure-Constrained, Observation-Evaluated Inventory of Oil and Gas Fugitive Emissions in Pakistan"

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Release provenance. This document describes PakEmis under the period, configuration and source boundaries specified in its methods. The identifiers below link it to the accompanying products and source register.

Release version	`1.0.0rc68`
Commit	`6d8e8669a460`
Reporting period	July 2015–June 2025 (ten fiscal years) (2015-07-01 to 2025-06-30)
Reference	S_TAN (household survey; TAN nitrogen)
FY2023–24 total	444.61 Mt CO_2e (AR5) under S_TAN, the reference; 449.85 Mt under L_TAN
Release manifest	`e6877cc36bb6604c` (SHA-256 of `outputs/release_manifest.json`, first 16)
Source manifest	`c1be6d314cfce1ea` (SHA-256 of `data/manifests/source_manifest.csv`, first 16)
Data-description article	`paper_national/` (ESSD, in preparation), same release

1. Observation-to-field match audit

Table S1 exposes every join used to connect a UNEP MARS source to the nearest World Bank flare and then to the canonical official field ledger. "High" requires an official field, distance no greater than 10 km, a known operator and a non-transmission source. Records at 10–25 km, transmission sources or records without operator metadata require review. A record is unmatched when the named flare field is absent from the 73-field official supply ledger or the distance exceeds 25 km. Unmatched observations remain in the source archive but do not enter named-field statistics. Coordinates, operator and flare identifier are retained in `outputs/fugitives_observation_match_audit_2024.csv`.

Table S1: Nearest-flare portion of the MARS match audit. Source identifiers have the common `PAK_S_` prefix.

ID	MARS source type	Flare field	Flare operator	km
002	transmission compressor	Bhit	Prime International Oil & Gas	0.07
003	transmission compressor	Naimat West	United Energy	0.35
013	gas disposal	Loti	OGDC	3.97
011	gathering/boosting	Sharf	Pakistan Petroleum	0.13
014	gathering/boosting	Bhit	Prime International Oil & Gas	1.20
015	O&G facility	Mari	Mari Petroleum Company	6.56
016	O&G facility	Pasakhi	OGDC	1.98
005	gas disposal	Loti	OGDC	3.99
006	gas disposal	Loti	OGDC	3.64
001	gathering/boosting	Mari	Mari Petroleum Company	0.07
010	gas-processing unit	Rajani	United Energy	0.22
009	flare	Sharf	Pakistan Petroleum	0.06
008	gathering/boosting	Qadirpur	OGDC	0.52
020	O&G source	Kadanwari	Prime International Oil & Gas	0.12
022	transmission station	Qadirpur	OGDC	2.69
023	gathering/boosting	Uch	OGDC	0.27
027	O&G facility	Bhit	Prime International Oil & Gas	14.06
024	gathering/boosting	Chanda	OGDC	0.20
026	gathering/boosting	Nashpa	OGDC	0.35
025	gathering/boosting	Manzalai–Kohat (2)	unreported	0.36

Table S2: Canonical-field decision and confidence audit.

ID	Canonical field	Method	Status	Reason
002	bhit	exact normalized	review	transmission source
003	naimat west	exact normalized	unmatched	absent official field; transmission
013	loti	exact normalized	unmatched	absent official field
011	sharf	exact normalized	unmatched	absent official field
014	bhit	exact normalized	high	—
015	mari	exact normalized	high	—
016	pasakhi	exact normalized	unmatched	absent official field
005	loti	exact normalized	unmatched	absent official field
006	loti	exact normalized	unmatched	absent official field
001	mari	exact normalized	high	—
010	rajani	exact normalized	unmatched	absent official field
009	sharf	exact normalized	unmatched	absent official field
008	qadirpur	exact normalized	high	—
020	kadanwari	exact normalized	high	—
022	qadirpur	exact normalized	review	transmission source
023	uch	exact normalized	high	—
027	bhit	exact normalized	review	distance >10 km
024	chanda	exact normalized	high	—
026	nashpa	exact normalized	high	—
025	manzalai	alias	review	operator missing

2. Sparse-field permutation evaluation

All 73 official fields and all zero-observation fields are retained. For each comparison, 10,000 deterministic seeded permutations shuffle observation values either across the whole field set or only among fields in the same province. Spearman tests are two-sided; top-10 overlap tests use the upper tail. The +1 correction is applied to numerator and denominator.

Observation	Null	Nonzero fields	ρ	p_ρ	Top-10 overlap	p_{top}
VIIRS flare volume	national	25	0.230	0.0514	4/10	0.0224
VIIRS flare volume	within province	25	0.230	0.0369	4/10	0.0257
MARS sources	national	8	0.382	0.0003	4/8	0.0065
MARS sources	within province	8	0.382	0.0013	4/8	0.0374
MARS plumes	national	8	0.382	0.0004	4/8	0.0093
MARS plumes	within province	8	0.382	0.0007	4/8	0.0370

Table S3: Seeded permutation diagnostics. Observation quantities are not annualized emissions.

3. Completeness and materiality

The screen threshold is 1% of the reference configuration (0.235 Mt CO_2e). No numerical upper bound is assigned where a defensible public activity is absent. “IE” denotes inclusion elsewhere in an inclusive factor; “NE” denotes not estimated; “NO” denotes reported non-occurrence; and “NA” denotes no applicable GHG factor in the 2019 Tier-1 table.

IPCC	Subsegment	Key	Materiality	Priority evidence
1.B.1	Coal mining	NE	out of scope/unknown	Coal output by mine type and mine-gas content
1.B.2.a.i	Oil exploration	NE	unknown	Exploration wells and testing events
1.B.2.a.ii	Oil production/upgrading	IE	estimated/included	Onshore/offshore split, active wells and practices
1.B.2.a.iii	Oil transport	E/partial	estimated/included	Mode-specific pipeline, truck, rail and tanker throughput
1.B.2.a.iv	Oil refining	E/proxy	estimated/included	Crude feed by refinery
1.B.2.a.v	Oil-product distribution	NA	not applicable	Add NMVOC to the pollutant inventory if required
1.B.2.a.vi	Oil other	NE	not applicable	Incident and spill records
1.B.2.a.vii	Abandoned oil wells	NE	unknown	Plugged/unplugged well census
1.B.2.b.i	Gas exploration	NE	unknown	Exploration wells and testing events
1.B.2.b.ii	Gas production/gathering	IE	estimated/included	Well counts, field throughput and practices
1.B.2.b.iii	Gas processing	E/proxy	estimated/included	Plant throughput, acid-gas removal and compressor seals
1.B.2.b.iv	Gas transmission	E	estimated/included	Independent pipes, LDAR and dry/wet seal shares
1.B.2.b.iv	Gas storage	NO	probably minor/NO	Reassess when an operating facility is reported
1.B.2.b.v	Gas distribution	E	estimated/included	Pipe material, pressure and LDAR coverage
1.B.2.b.vi	Post-meter appliances	E/proxy	estimated/included	Appliance census and commercial multiplier
1.B.2.b.vi	Natural-gas vehicles	NE	unknown	Active CNG-vehicle count by technology
1.B.2.b.vi	Industrial/power post-meter	E	estimated/included	Facility-specific post-meter measurements
1.B.2.b.vii	Other/accidents	NE	potentially material	Rupture, blowdown and super-emitter records
1.B.2.c	Venting/flaring	IE/memo	estimated/included	Operator vented/flared gas and Annex 4A.2 split

Table S4: Complete notation-key and materiality matrix.

4. Material uncertainty-input register

All entries use bounded split-triangular sampling in the reference analysis; the deterministic value is the mode and median. Published IPCC bounds are treated as expert/table ranges, not as fitted confidence intervals. Shared activity-source deviates are applied once across connected segments, while additional proxy-coverage residuals remain independent. The complete register, including minor oil, LNG, appliance, CO_2 and N_2O rows and zero-bound handling, is archived in `outputs/fugitives_uncertainty_inputs_2024.csv`.

Input/group	Central	Lower	Upper	Unit	Shared with	Source
Gas production activity	32309	30694	33925	MMm^3	production/gather/process	Economic Survey
Gas-processing activity	32309	25848	38771	MMm^3	production + proxy residual	activity proxy
Sour-processing activity	10339	8271	12407	MMm^3	production + proxy residual	32% default/proxy
Distribution pipeline	201230	197205	205255	km	distribution gases	OGRA
Post-meter activity	18685	17750	19619	MMm^3	post-meter gases	OGRA end uses
Gas-production CH_4 EF	4.09	3.272	4.908	t/MMm^3	production gases	IPCC 4.2.4g
Gas-processing CH_4 EF	1.83	1.647	2.013	t/MMm^3	processing gases	IPCC 4.2.4h
Sour-processing CO_2 EF	66.6	0	133.2	t/MMm^3	sour-processing gases	IPCC 4.2.4h
Distribution CH_4 EF	1.17	0.936	2.574	t/km	distribution gases	IPCC 4.2.4j
Post-meter CH_4 EF	6.1	2.44	9.76	t/MMm^3	post-meter gases	IPCC 4.2.4k

Table S5: Material reference-configuration inputs.

5. Conditional uncertainty and distribution shape

For reported value x, proportional lower and upper ranges l and h, and $u \sim \mathcal{U}(0,1)$, the bounded split-triangular multiplier is

$$m(u) = \begin{cases} 1 - l + l\sqrt{2u}, & u < 0.5, \\ 1 + h - h\sqrt{2(1-u)}, & u \geq 0.5. \end{cases} \tag{1}$$

The reported input is the mode and median. Gas production, gathering and processing share the production-activity draw; additional processing-coverage residuals are independent. Oil production and its transport fallback share the oil draw. Gases belonging to one physical factor group share a quantile transformed through their gas-specific bounds. All simulations use 10,000 iterations and seed 42.

Experiment	Distribution	Deterministic	P2.5	Median	P97.5
Reference	split triangular	23.499	21.277	24.141	30.537
Central (geometric-mean practice)	split triangular	17.678	15.798	18.137	21.309
Lower practices	split triangular	14.894	13.113	15.144	17.311
No industrial post-meter	split triangular	20.079	18.489	20.526	26.836
Minimum coverage	split triangular	9.601	8.864	9.745	10.692
Reference shape test	bounded uniform	23.499	21.351	27.140	33.140
Reference shape test	bounded log-uniform[a]	23.499	20.300	25.239	31.909

Table S6: Conditional 95% intervals for the five interpretable configurations, and distribution-shape sensitivity, Mt CO_2e. The central member is a synthesized geometric-mean reading, priced and never selected.

[a]Inputs with a zero lower bound use bounded uniform sampling.

6. Official-inventory and computational audit

The CRT audit preserves the source ZIP and workbook hashes, worksheet names, cell coordinates, raw values, units, formulas and notation keys. Generated tables reproduce each numeric activity–factor row in native gases, convert the submitted total to the common GWP basis and close every step of the CRT-to-reconstruction waterfall. The complete records are archived in `outputs/fugitives_unfccc_crt_cell_audit.csv`, `outputs/fugitives_unfccc_crt_reproduction.csv` and `outputs/fugitives_unfccc_crt_waterfall_2022.csv`.

Automated QA covers factor units, registered activity years, source-row and segment closure, pipeline and gas-flow identities, all 280 structural configurations, conditional interval ordering, field-supply reconciliation, observation-match completeness, permutation reproducibility, provincial reconciliation and manuscript-output agreement. The run manifest and machine-readable result are stored in `outputs/fugitives_run_manifest_2024.json` and `outputs/fugitives_qa_2024.csv`.